**Ramrao Adik Institute of Technology, Nerul, Navi Mumbai**

Module 1

Academic year: 2026-2027

Subject: Embedded System Class: S.E. Sem : III

1. Convert $(73.301)_{10}$ into binary, octal and Hexadecimal equivalent.
2. Convert the following.
   i) $(BC)_{16} = (\ )_{10}$
   ii) $(2314)_{8} = (\ )_{10}$
   iii) $(1000011)_{2} = (\ )_{10}$
   iv) $(647)_{10} = (\ )_{16}$
3. Express the following numbers in decimal.
   i) $(10110.0101)_{2}$
   ii) $(16.5)_{16}$
   iii) $(26.24)_{8}$
4. Give the truth table and symbol of basic logic gates.
5. Give the truth table and symbol of Universal logic gates.
6. Give the truth table and symbol of exclusive logic gates.
7. Design & implement the Half Adder using logic gates.
8. Design & implement the Full Adder using logic gates.
9. What is Multiplexer? Draw a 4:1 Multiplexer, write down the Boolean equation and design it with logic gates.
10. Explain the working of 4:1 Demultiplexer. Design it with logic gates.
11. Draw the circuit of SR Latch and explain its working with truth table.
12. Draw the circuit of D- flip flop and explain its working with truth table.
13. Draw the circuit of SR flip-flop and explain its working with truth table.

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Module - 2

Academic year: 2026-2027

Subject: Embedded System Class: S.E. Sem : III

1. What is an embedded system? How does it differ from a general-purpose computer?
2. Explain the basic classification of Embedded Systems.
3. Describe the applications of Embedded Systems.
4. What are the different characteristics of Embedded Systems? Explain in detail.
5. What are the key components of an embedded system, and how do they interact?
6. List and explain different features of Embedded Cores.
7. Differentiate RISC and CISC architecture.
8. What is ASIC? Explain briefly.
9. Compare Microprocessors and Microcontrollers.
10. What is the Real Time Embedded System? Also explain what you mean by Hard Real time systems and Soft real time systems with suitable example.
11. What are the different memory types used in embedded systems?
12. Explain the concept of power management in embedded systems. Why is it important?

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Module-3

Academic year: 2026-2027

Subject: Embedded System Class: S.E. Sem : III

1. Draw the architecture of 8051 and explain in detail.
2. Draw the block diagram of 8051 and explain the main blocks.
3. Draw the pin-out of 8051 and explain the function of all pins.
4. Explain the arithmetic and logical type of instructions of 8051 with examples.
5. Explain data transfer type of instructions of 8051 with example.
6. Explain the bit manipulation type of instructions of 8051 with examples.
7. Explain branching type of instructions in 8051 with examples.
8. Explain the RET and RETI instructions of 8051 with suitable examples.
9. Explain which instruction in 8051 is used for creating a loop in program.
10. Explain all types of jump instructions in 8051 with suitable examples.
11. Write a Assembly Language Program in 8051 for Addition of two numbers.
12. Write a Assembly Language Program in 8051 for Subtraction of two numbers.
13. Write a Assembly Language Program in 8051 for Multiplication of two numbers.
14. Write a Assembly Language Program in 8051 to transfer a block of five numbers.

**Ramrao Adik Institute of Technology, Nerul, Navi Mumbai**

Module-4

Academic year: 2025-2026

Subject: Embedded System Class: S.E. Sem : III

1. List and explain all addressing modes in 8051 microcontroller.
2. Explain how I/O ports are initialized in 8051 with suitable instructions and examples.
3. Explain MUL and DIV instructions in 8051 with suitable examples.
4. Explain ANL, ORL and XRL instructions in 8051 with suitable examples.
5. Explain RL, RLC, RR, RRC instructions in 8051 with suitable examples.
6. Explain TMOD and TCON SFR's of 8051 microcontroller in detail.
7. Explain how 8051 can be used in Counter Mode.
8. Explain how 8051 serial communication works with suitable SFR's.
9. Explain what RS232 is and hence explain DTE and DCE too.
10. Explain the 8051 interrupts with Interrupt Vector table (IVT) in detail.
11. Write a assembly language program to find the largest number from given five numbers.
12. Write a assembly language program to unpack a packed BCD Number.
13. Write a assembly language program to generate a delay of 1ms.
14. Write a program for the 8051 to transfer letter "A" continuously on the serial port.
15. Write a program to detect the position of switch connected to P2.7 pin and accordingly ON and OFF the LED connected at pin P1.0. LED should be ON when the switch is in ON position and LED should be OFF when the switch is in OFF position.

**Ramrao Adik Institute of Technology, Nerul, Navi Mumbai**

Module-5

Academic year: 2026-2027

Subject: Embedded System Class: S.E. Sem : III

1. Draw detailed diagram of Seven Segment Display (SSD) interfacing with 8051 microcontroller and write suitable program for the same.
2. Discuss in detail the Embedded case study of “Digital Camera”.
3. Discuss in detail the Embedded case study of “Washing Machine”.
4. Draw detailed diagram of DC Motor interfacing with 8051 microcontroller and write suitable program for the same.
5. Draw detailed diagram of Stepper Motor interfacing with 8051 microcontroller and write suitable program for the same.
6. Draw detailed diagram of Liquid Crystal Display (LCD) interfacing with 8051 microcontroller and write suitable program for the same.
7. Draw detailed diagram of Hex Keyboard interfacing with 8051 microcontroller and write suitable program for the same.
8. Explain what is “Wave Drive, Full Step and Half Step” sequence with reference to stepper motor interfacing.
9. Assume that switch SW is connected to the pin P2.0 and P1.0 is connected to one of the input of DC Motor and P1.1 is connected to another input of DC Motor. Write a program to monitor the status of SW, if SW = 1, rotate the dc motor in a clockwise direction, or if SW = 0, reverse the direction of the motor.
10. Explain the difference between Common Cathode and Common Anode SSDs.

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Module-6

Academic year: 2026-2027

Subject: Embedded System Class: S.E. Sem : III

1. Explain the main features of ARM architecture.
2. Draw the ARM architecture and discuss the main blocks.
3. Discuss the General-Purpose Register set and CPSR in ARM architecture.
4. Describe the different operating modes in ARM architecture.
5. Explaining how pipelining process in ARM architecture works?
6. Write a short note on Cortex processors from the family of ARM CPU cores.
7. Elaborate the features of Cortex ARM processors.
8. Explain three stage pipelining process in ARM architecture.
9. Explain what Cortex-A, Cortex-R and Cortex-M in ARM architecture is.
10. What are the general features of Cortex ARM processors?